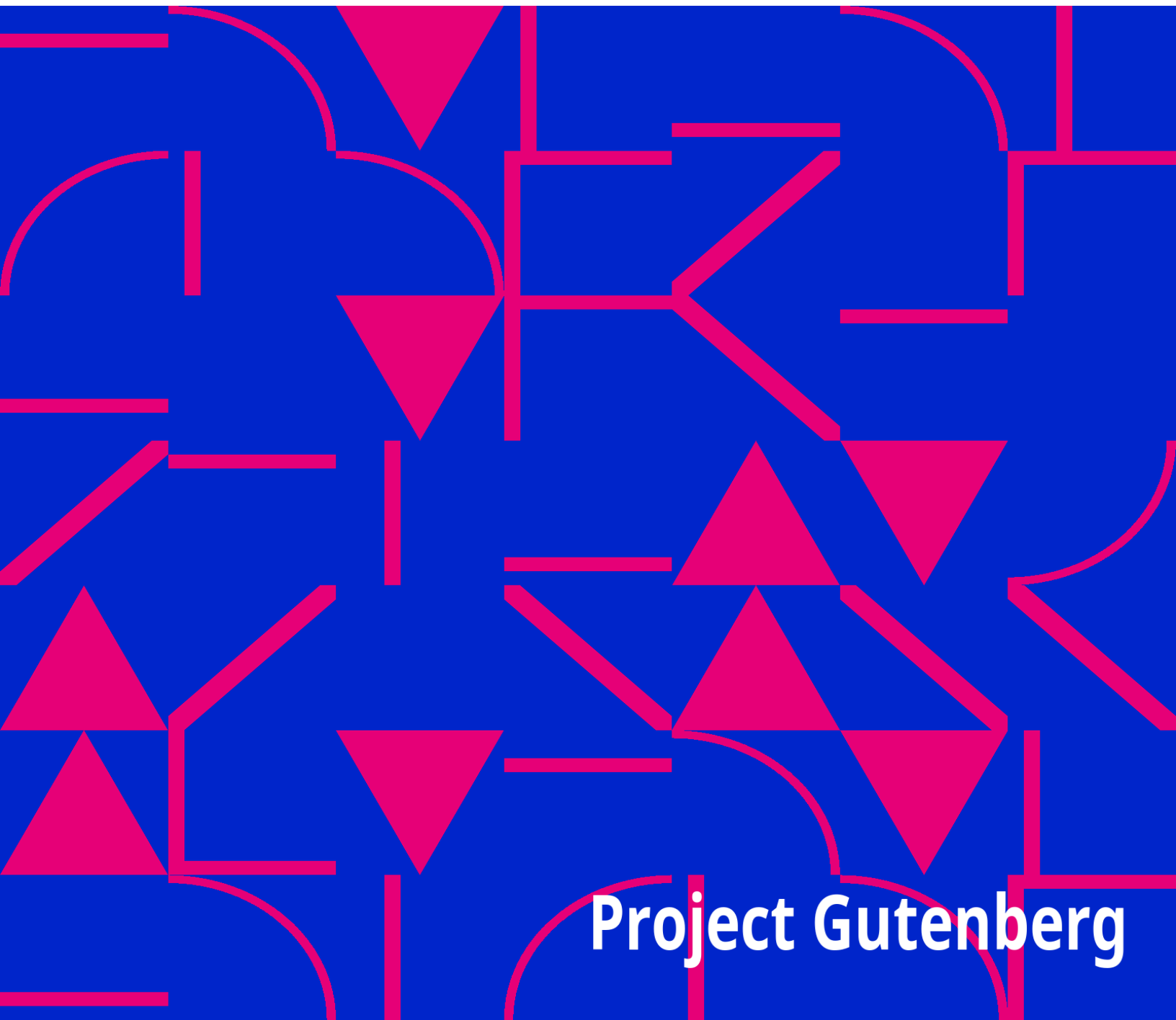


Distribution of Some Nebraskan Mammals

J. Knox Jones

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BY
J. KNOX JONES, JR.

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Distribution of Some Nebraskan Mammals

by

J. Knox Jones, Jr.

Because military service will interrupt my study of Nebraskan mammals, I am here placing on record certain information on the geographic distribution of several species—information that is thought pertinent to current studies of some of my associates. Most of this information is provided by specimens recently collected by me and other representatives of the University of Kansas Museum of Natural History, although specimens from other collections provide some of the records herein reported. The other collections are the Biological Surveys Collection of the United States National Museum (USBS), the Hastings Museum (HM), the Nebraska Game, Forestation and Parks Commission (NGFPC), the University of California Museum of Vertebrate Zoology (MVZ), the University of Michigan Museum of Zoology (MZ) and the University of Nebraska State Museum (NSM). Grateful acknowledgment hereby is made to persons in charge of these several collections for lending the materials concerned. Specimens mentioned in the following accounts are in the University of Kansas Museum of Natural History, except as otherwise stated. All measurements are in millimeters. Color terms are those of Ridgway (1912). A part of the funds for field work was made available by the National Science Foundation and the Kansas University Endowment Association.

Sorex cinereus haydeni. (Baird). CINEREOUS SHREW.—Two male shrews were trapped on April 7, 1952, among rocks along an old railroad fill, 4 mi. N, 1/2 mi. E of Octavia, Butler County, thus extending the known geographic range of *S. c. haydeni* approximately 60 miles southward from a line connecting Perch, Rock County, Nebraska, with Wall Lake, Sac County, Iowa (see Jackson, 1928:52-53), and providing the first record of occurrence in the Platte River Valley. Two additional specimens, taken on

July 17, 1952, are from 2-1/2 mi. N of Ord, Valley County, along the Loup River, a tributary of the Platte from the north.

Blarina brevicauda carolinensis (Bachman). SHORT-TAILED SHREW.—J. S. Findley and I, in a forthcoming paper, review the distribution of *Blarina brevicauda* in the Great Plains region, recording *B. b. carolinensis* from the extreme southeastern and southwestern counties of Nebraska. A series of five shrews of this species recently obtained from three miles south and two miles east of Nebraska City in Otoe County, average significantly smaller in both the cranial and the external measurements than typical *B. b. brevicauda* and fall well within the range of *carolinensis*. Average and extreme external measurements of the four adults from Otoe County, three males and one female, are as follows: Total length, 110 (109-112); length of tail-vertebrae, 24.2 (22-26); length of hind foot, 13.8 (13-14). Another specimen from 3 mi. S, 1-1/2 mi. E of Peru, Nemaha County, also is referable to *carolinensis*. These recent records indicate that the range of *B. b. carolinensis* extends up the Missouri River Valley, approximately to Nebraska City, Otoe County. Five specimens from Louisville, Cass County, the next county northward, along the River, are referable to *B. b. brevicauda*.

Eptesicus fuscus fuscus. (Beauvois). BIG BROWN BAT.—One big brown bat was obtained on July 23, 1952, from one mile west of Niobrara, Knox County. While not so dark in dorsal coloration as some specimens of *E. f. fuscus* from eastern Nebraska (Cass and Sarpy counties), this specimen is noticeably darker than a series of *E. f. pallidus* from Ft. Niobrara Wildlife Refuge, 4 mi. E of Valentine, Cherry County, being near (16" j) Snuff Brown as opposed to near (16' i) Buckthorn Brown. Previous to the taking of this specimen, Webb and Jones (1952:277) reported as *E. f. pallidus* a specimen, saved as a skull only, which was picked up dead at Niobrara. It seems best to assign these two bats from the vicinity of Niobrara, Knox County, to *E. f. fuscus*.

Sciurus carolinensis carolinensis Gmelin. GRAY SQUIRREL.—An adult male gray squirrel shot by Mr. Terry A. Vaughan in the heavily timbered bluffs of the Missouri River, 3 mi. S, 2 mi. E of Nebraska City, Otoe County, on October 10, 1953, provides the only museum specimen of a gray squirrel from Nebraska known to me. Residents in the area concerned report small

numbers of this squirrel as still occurring on the heavily wooded bluffs along the Missouri River in Nemaha, Otoe and Richardson counties, Nebraska, at least as far north as Nebraska City.

Gray squirrels from Nebraska have been reported twice before in the literature as follows: "Mouth of Platte [River]" (Baird, 1858:262) and Barada, Richardson County (Jones and Webb, 1949:312). Swenk (1908:80), while listing no actual records, says of this squirrel, "Common in the timber along watercourses of southeastern Nebraska, but greatly outnumbered everywhere by [*Sciurus niger*] *rufiventer*. I have no records west of the 97th meridian nor north of the Platte."

Spermophilus franklinii (Sabine). FRANKLIN GROUND SQUIRREL.—A specimen from 2 mi. NW of Lisco, in Morrill County (NSM 3324), extends the known geographic range of *S. franklinii* approximately 200 miles westward along the Platte River Valley from Kearney, Buffalo County (see Howell, 1938:134), and suggests a westward movement of this ground squirrel along the Platte River in recent years.

Perognathus flavescens flavescens Merriam. PLAINS POCKET MOUSE.—*P. f. flavescens* occurs in the Sand Hills and adjacent mixed-grass plains of central Nebraska. Eastern marginal records of occurrence are: Neligh, Antelope County, 2 (MVZ 1, NSM 1); 1 mi. E of Ravenna, Buffalo County, 2 (MZ); unspecified locality in Adams County, 1 (HM).

Perognathus flavescens perniger Osgood. PLAINS POCKET MOUSE.—This mouse occurs in northeastern Nebraska. Osgood (1904:127), in the original description of the subspecies, listed two specimens from Verdigris [Verdigre], Knox County. Additional records of occurrence are: Beemer, Cuming County, 2 (USBS); 1-1/2 mi. SE of Niobrara, Knox County, 3; 1-1/2 mi. S of Pilger, Stanton County, 2.

The two specimens from Beemer are typical *perniger*. All of the other Nebraskan specimens are intergrades between *P. f. flavescens*, geographically adjacent to the west, and *P. f. perniger* to the east but are best referred to *perniger* on the basis of greater total length, larger cranial measurements and darker dorsal coloration.

P. f. perniger was originally described (Osgood, *op. cit.*) on the basis of its darker dorsal coloration and encroachment of the lateral line on the posterior parts of the venter. The latter character is not present in all Nebraskan specimens. Mice from the two localities in Knox County have buffy underparts; those from other Nebraskan localities do not. Of nine specimens of *P. f. perniger* examined from Elk River, Sherburne County, Minnesota, none has buffy underparts whereas a specimen from Randolph, Fremont County, Iowa (NSM) does. In addition, in two of five specimens of *P. f. flavescens* from Kelso, Hooker County, (MZ) the lateral line encroaches on the underparts. The encroachment of the lateral line on the underparts, or failure of the line to do so, is thought to be only an individual variation and of no taxonomic use.

Perognathus flavus piperi Goldman. BUFFY POCKET MOUSE.—In the description of *P. f. bunkerii*, Cockrum (1951:206) allocated to the new subspecies, without comment, a specimen from Alliance, Box Butte County. I have examined this specimen along with all other Nebraskan specimens known to me and, although all approach *bunkerii* in cranial measurements, they seem best referred to *piperi* on the basis of darker dorsal coloration and larger external measurements. Additional records of occurrence, several of them marginal to the eastward, are: 10 mi. S of Antioch, Garden County, 1 (MZ); Kelso, Hooker County, 4 (MZ); 5 mi. N of Bridgeport, Morrill County, 1 (MVZ); 6 mi. N of Mitchell, Scotts Bluff County, 1 (NSM). A specimen not seen by me that was reported from Valentine, Cherry County (Beed, 1936:21), is presumably also best referred to *P. f. piperi*.

No specimens of *P. flavus* are known to me from south of the Platte River in southwestern Nebraska although they probably occur there. If so, they may be referable to *P. f. bunkerii*, which is found in counties of Kansas adjoining the southwestern part of Nebraska.

Perognathus hispidus paradoxus Merriam. HISPID POCKET MOUSE.—This subspecies occurs commonly in central-and western-Nebraska. Eastern marginal records of occurrence are: 2 mi. SE of Niobrara, Knox County, 1 (NGFPC); 4 mi. E, 2 mi. S of Ord, 1; Bladen, Webster County, 2 (HM).

Perognathus hispidus spilotus Merriam. HISPID POCKET MOUSE.—Jones and Webb (1949:312) first reported this subspecies in Nebraska as from 5

mi. SE of Rulo, Richardson County. Additional records of occurrence are: 3 mi. SW of Barnston, Gage County, 1 (NGFPC); Bennet, 1 (NSM), 9 mi. NW of Lincoln, 1 (NSM), 1-1/2 mi. S of Lincoln, 1 (NSM), Lancaster County; Peru, Nemaha County, 1 (NGFPC); 3 mi. S, 2 mi. E of Nebraska City, Otoe County, 3; Barada, Richardson County, 1 (NSM); Pleasant Dale, Seward County, 1 (NSM); 1 mi. S of Williams, Thayer County, 1.

Glass (1947:179) referred a specimen from 9 mi. NW of Lincoln, Lancaster County, to *P. h. paradoxus*. In discussing the zone of intergradation between *spilotus* and *paradoxus*, geographically adjacent to the west, he wrote (*op. cit.*:178), "It is evident that it proceeds northeastwards, toward the Missouri River since 2 specimens from eastern Nebraska, a juvenile from Webster County and an adult from Lancaster County, are both typical *paradoxus*." I have examined the specimen from Webster County referred to by Glass and agree that it is *paradoxus*. I have not seen the specimen from 9 mi. NW of Lincoln; however, another specimen from there, two others from Lancaster County, and one from Seward County (see above), are here referred to *P. h. spilotus*, rather than *P. h. paradoxus*, on the basis of notably darker dorsal coloration and smaller external and cranial measurements. The range of *P. h. spilotus* in Nebraska, as presently known, therefore, is limited to the eastern, more humid part of the State, south of the Platte River.

Peromyscus maniculatus osgoodi Mearns. DEER MOUSE.—Swenk (1908:95) reported this subspecies, under the name *Peromyscus nebrascensis*, from Glen, and Dice (1941:17) reported the subspecies from Agate, both localities being in Sioux County in the northwestern part of the State. Osgood (1909), however, did not mention Nebraskan specimens of this subspecies and excluded it from the State on his (*op. cit.*) distribution map of the subspecies of *P. maniculatus*. In addition, Quay (1948:181) reports, as *P. m. nebrascensis*, deer mice obtained by him in the badlands of northern Sioux County and adjacent Niobrara County, Wyoming. Four deer mice referable to *P. m. osgoodi* have been obtained from several localities on the Pine Ridge in Dawes County as follows: 3 mi. E of Chadron, 2; Chadron State Park, 1; 3 mi. SW of Crawford, 1. When compared with specimens of *P. m. nebrascensis*, geographically adjacent to the east, these mice are seen to be notably darker and less buffy than *nebrascensis* and to average significantly larger in both external and cranial measurements. All deer mice from the Pine Ridge and adjacent badlands of extreme

northwestern Nebraska probably are best referred to *P. m. osgoodi*. External measurements of two adult females are respectively: Total length, 180, 175; length of tail-vertebrae, 78, 74; length of hind foot, 19, 20; length of ear, 17, 16.

Neotoma floridana campestris J. A. Allen. FLORIDA WOOD RAT.—Five wood rats from 5 mi. N, 2 mi. W of Parks, Dundy County, in extreme southwestern Nebraska, provide the first record of occurrence of this subspecies in Nebraska. These animals were trapped in outlying sheds at the Rock Creek State Fish Hatchery. Two large wood-rat houses were in a dense thicket of brush and young trees in a small draw on the west side of the most westwardly hatchery lake. Brown rats (*Rattus norvegicus*) inhabited a combination garage-storage barn at the hatchery and no wood rats were taken there.

Microtus pennsylvanicus pennsylvanicus (Ord). PENNSYLVANIA MEADOW MOUSE.—This subspecies occurs in eastern and central Nebraska (see Bailey, 1900:18 and Swenk, 1908:104). Additional records of occurrence are as follows: 5 mi. E of Rising City, Butler County, 5; 4 mi. SE of Laurel, Cedar County, 1; Wayne, 2, and 2-1/2 mi. E of Wayne, 1, Wayne County; 2-1/2 mi. N of Ord, Valley County, 4.

Synaptomys cooperi gossii (Coues). COOPER LEMMING MOUSE.—Fichter and Hanson (1947:1-8) reported the first known occurrence of this microtine in Nebraska, recording specimens from several localities in Lancaster County and one from near Valentine, Cherry County. Recent records of this mouse which help to clarify its distribution in Nebraska are as follows: 4 mi. N, 1/2 mi. E of Octavia, Butler County, 1; 5 mi. N, 2 mi. W of Parks, Dundy County, 1; 1 mi. N of Pleasant Dale, Seward County, 1.

An adult female from Dundy County provides the westernmost record of distribution of the species in North America. The animal was trapped on November 1, 1952, in association with *Microtus pennsylvanicus modestus* in a marshy area at the Rock Creek State Fish Hatchery on spring-fed Rock Creek. The pelage on the back is notably darker than in *S. c. gossii*, and resembles *S. c. paludis* from the Cimarron River drainage in Meade County, Kansas, but in the sum total of its characters it most closely resembles *S. c. gossii* among named subspecies.

Mustela rixosa campestris Jackson. LEAST WEASEL.—The least weasel occurs in eastern and central Nebraska (see Swenk, 1926:313-330 and Hall, 1951:192) but is known by only a single specimen from each locality of record save for the area around Inland, Clay County (Swenk, *op. cit.*). Additional records of the distribution of this mustelid in Nebraska are: Hastings, Adams County, 1 (HM); Schuyler, Colfax County, 1 (NGFPC); Goehner, Seward County, 1 (NSM); 10 mi. S of Ord, Valley County, 1 (NGFPC). The last mentioned specimen, a skull only, was obtained from a pellet of an unidentified raptorial bird.

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